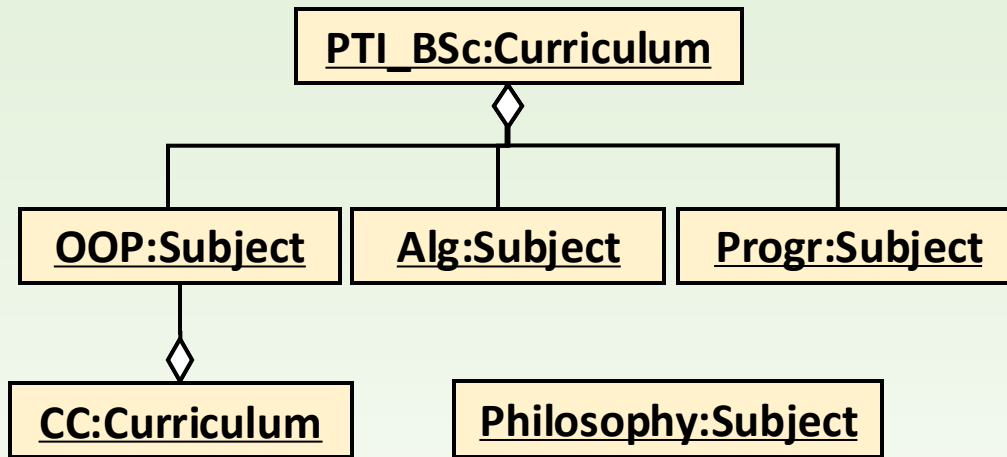
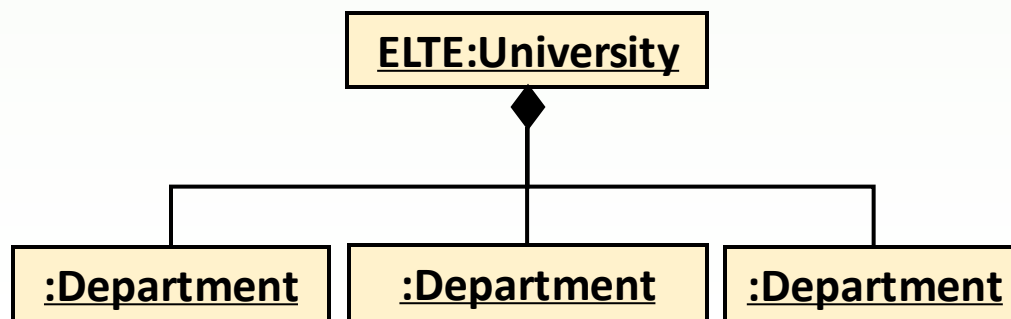


1/a A curriculum consists of subjects. A subject can be part of multiple curricula at the same time, and it does not have to be part of any curriculum.



1/b. A university has departments: there is no university without departments, and no department exists without a university. A department can only belong to one university and ceases to exist if its university is closed.



aggregation:

The same object can have multiple owners, and an owner can exist without an owned object

Curriculum

A subject can belong to 0, 1, ... curricula

Subject

Curriculum needs at least one subject

If multiplicity is not specified, it defaults to 1

composition: the container cannot exist without the contained, the contained can only be a part of one container

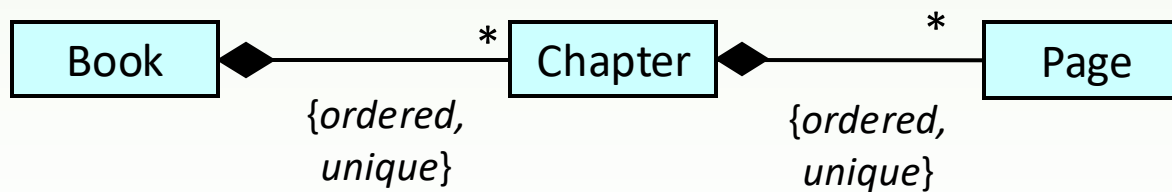
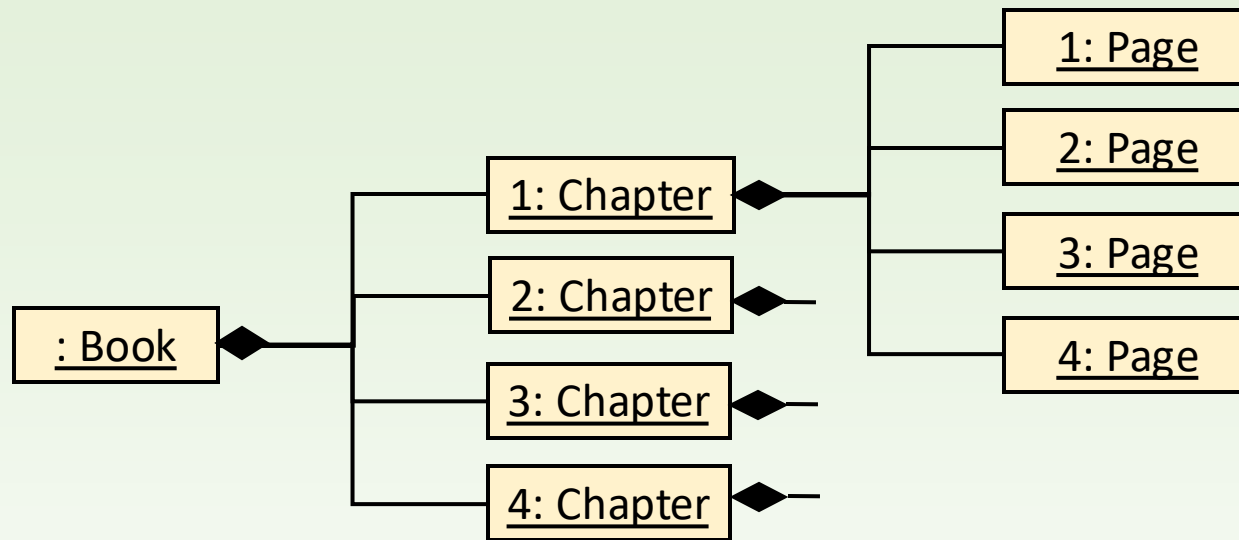
University

1 or 0..1 because of composition

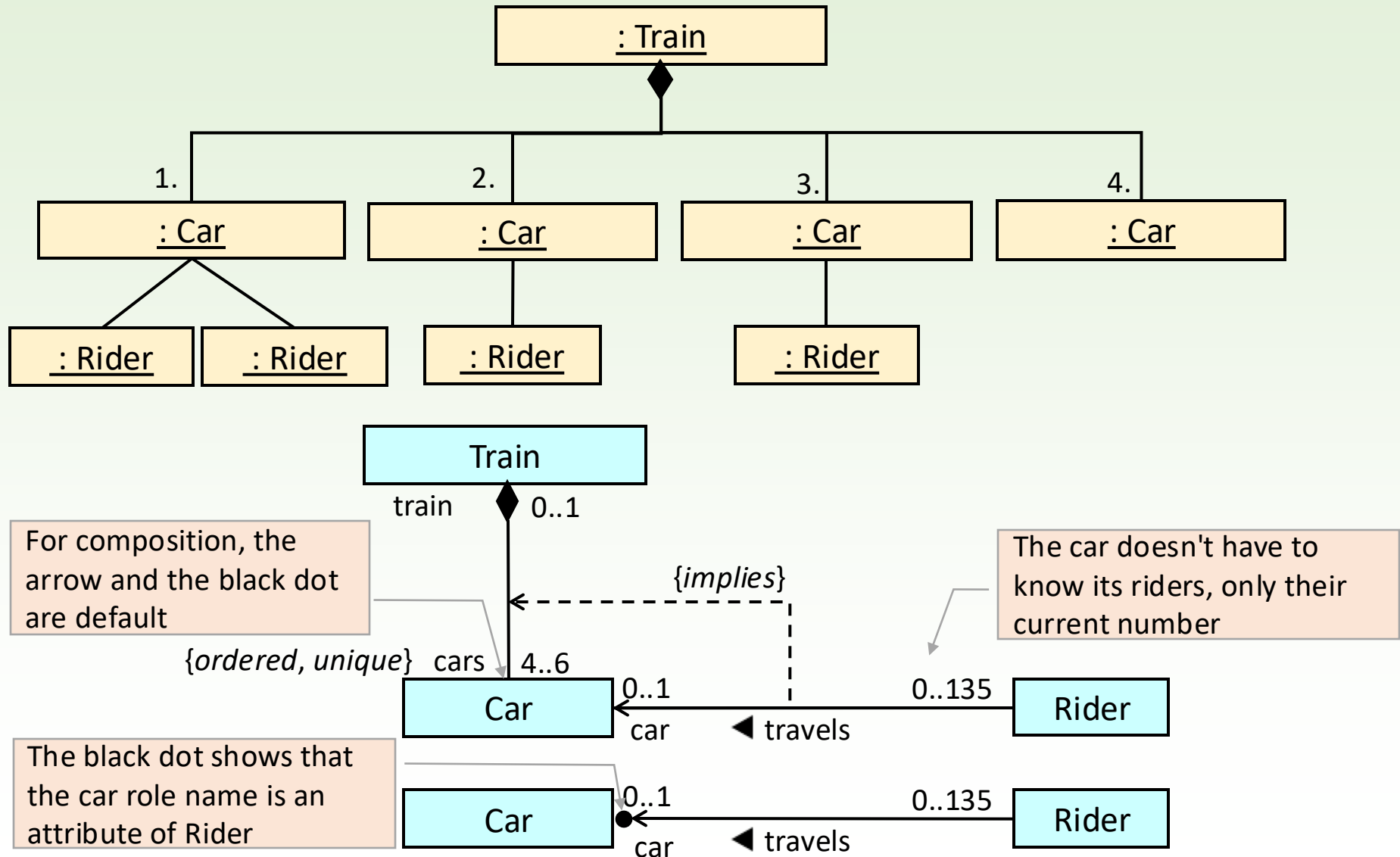
Department

This means 1..* because of the composition

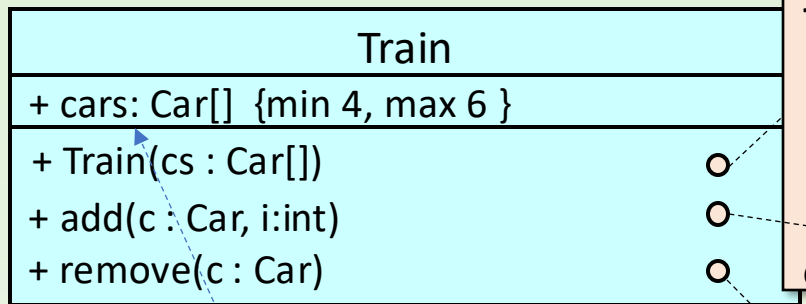
2. A book consists of at least one chapter, and a chapter contains at least one page.



3. A metro train consists of four, five, or six cars connected in a specific order. A passenger can board a metro car only if it is already part of a train. A metro car can accommodate up to 135 passengers.



Methods needed for relationships



```

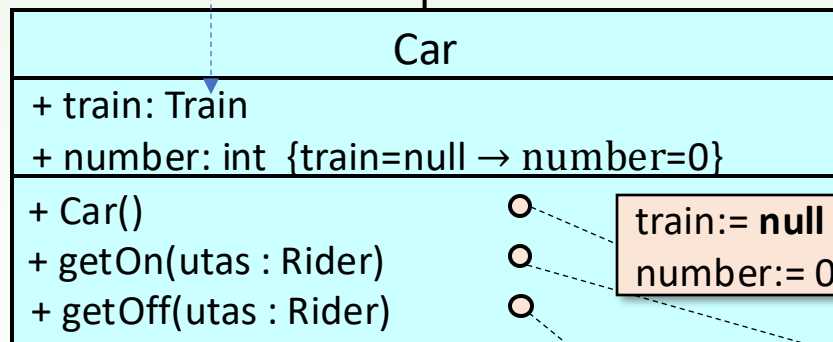
if |cs|<4 or |cs|>6 then error endif
forall c in cs loop
  if c.train≠null then error endif
  if c in cars then error endif
  cars.add(c)
  c.train:=this
endloop
  
```

```

if |cars|=6 then error endif
if c.train≠null then error endif
if c in cars then error endif
cars.insert(i, k)
c.train:=this
  
```

```

if |cars|=4 then error endif
if c.number>0 then error endif
cars.remove(c)
c.train:=null
  
```



```

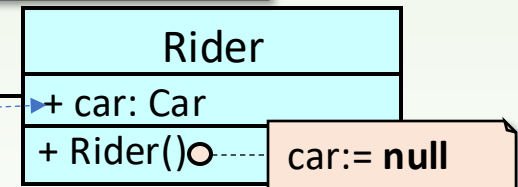
train:= null
number:= 0
  
```

```

if train= null then error endif
if number= 135 then error endif
if rider.car≠ null then error endif
number:= number+1
rider.car:=this
  
```

```

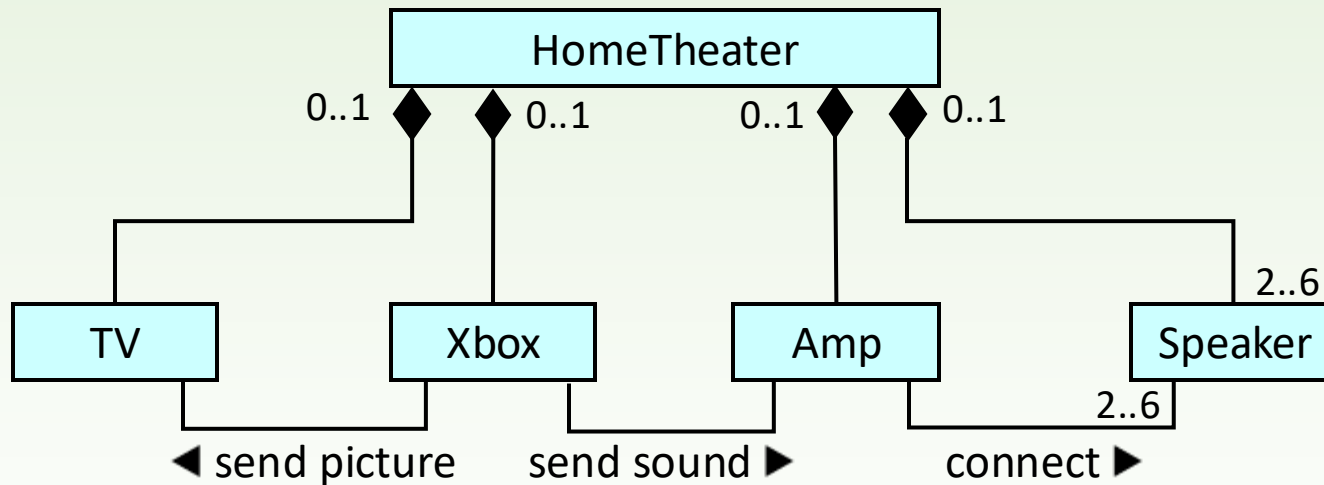
if rider.car≠ this then error endif
number:= number-1
rider.car:=null
  
```



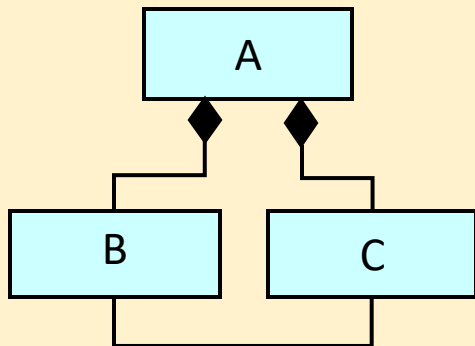
```

car:= null
  
```

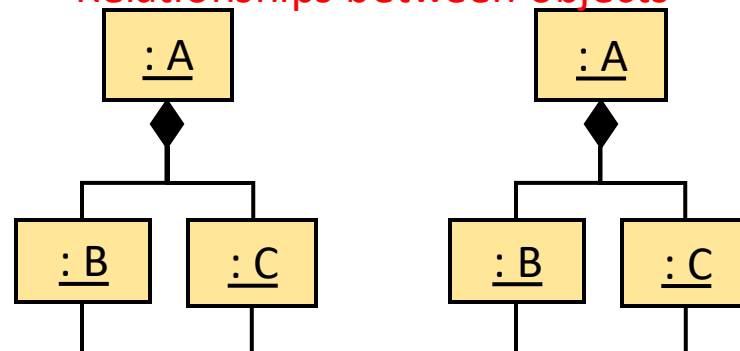
4. A home theater setup includes a TV, an Xbox, an amplifier, and the speakers. The number of speakers can range from two to six. The Xbox sends the video signal to the TV and the audio signal to the amplifier. The amplifier connects the speakers.



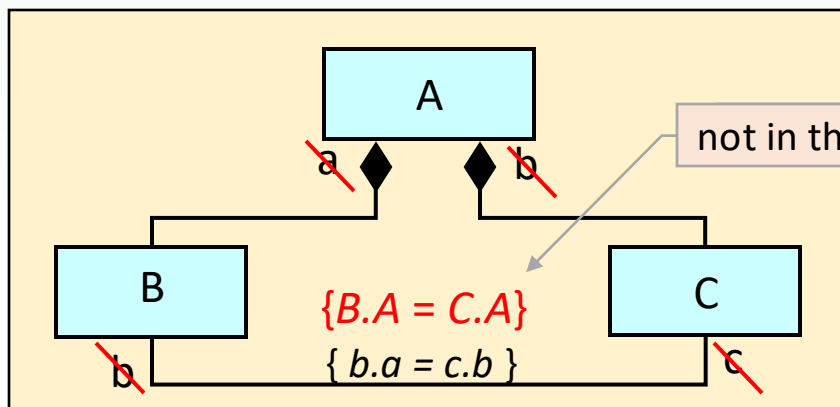
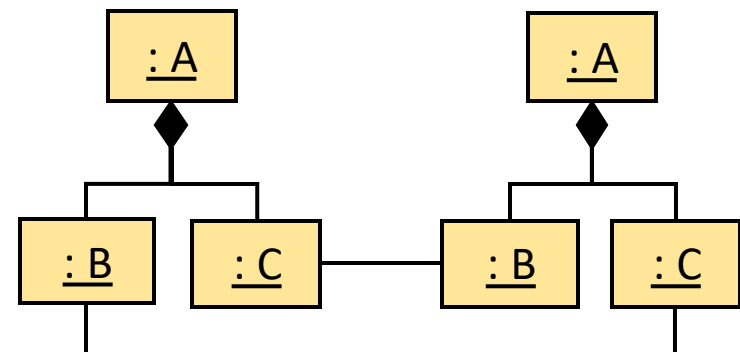
Model



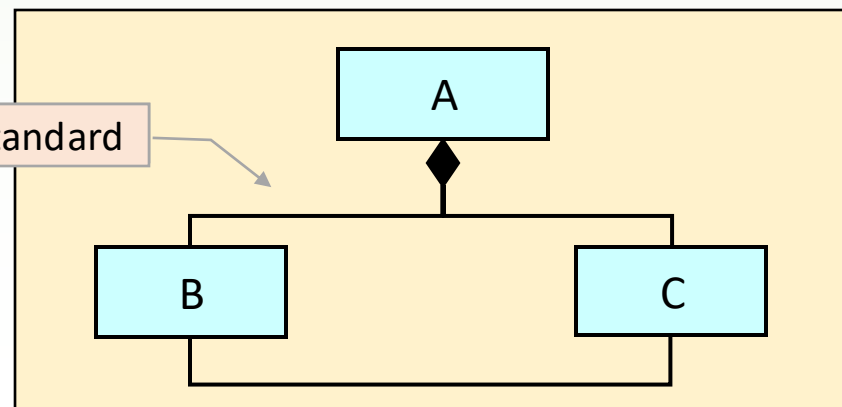
Relationships between objects



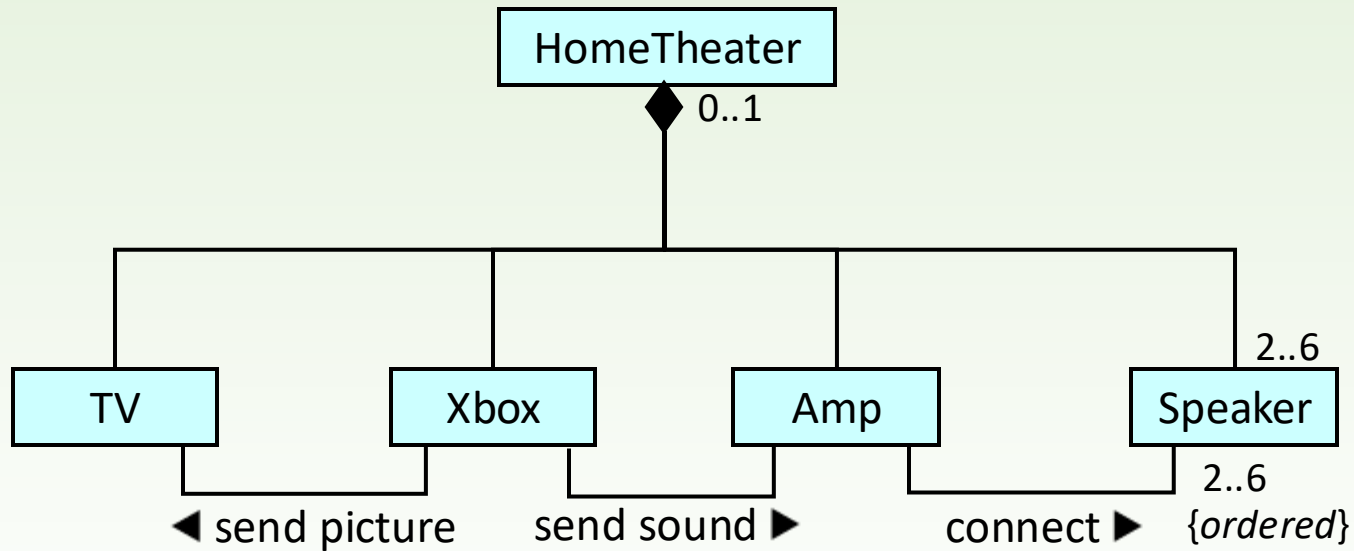
But this also conforms to the model



≡

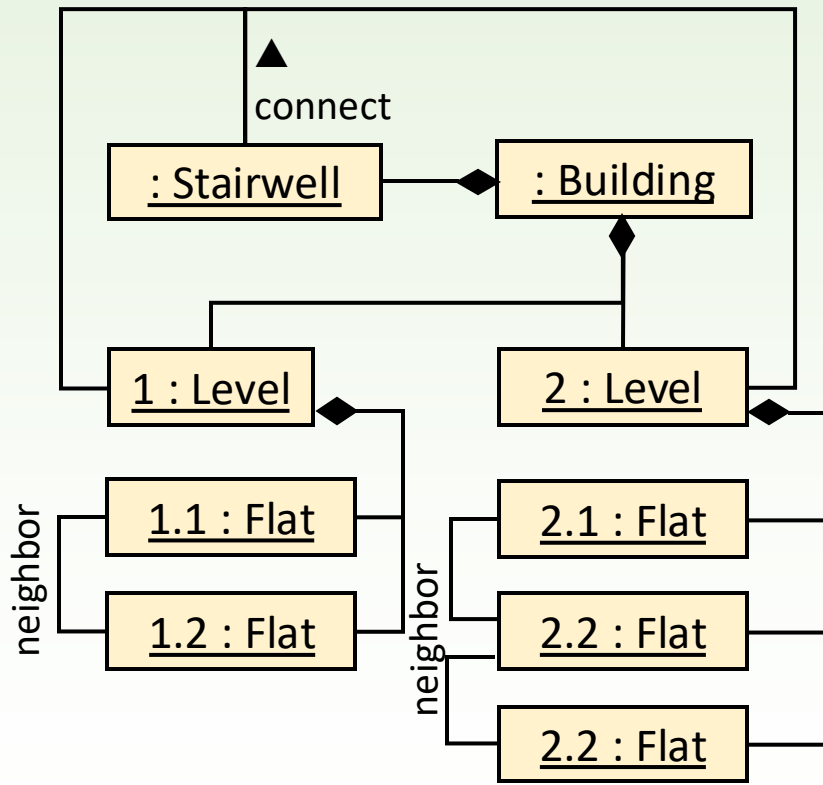


4. A home theater setup includes a TV, an Xbox, an amplifier, and the speakers. The number of speakers can range from two to six. The Xbox sends the video signal to the TV and the audio signal to the amplifier. The amplifier connects the speakers.

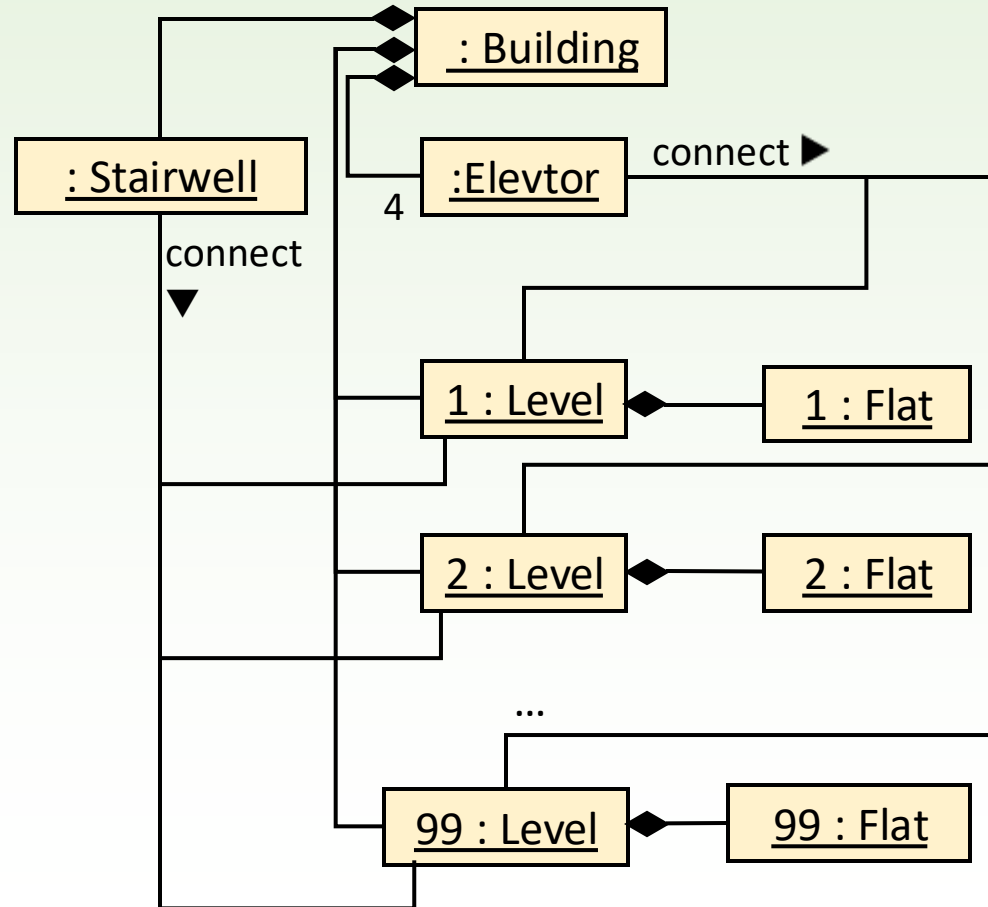


5. A residential building consists of at least two levels, with at least one apartment (flat) per level. The levels are connected by stairwells and elevators, with each level having at least one stairwell. Each apartment can have no more than two neighboring apartments.

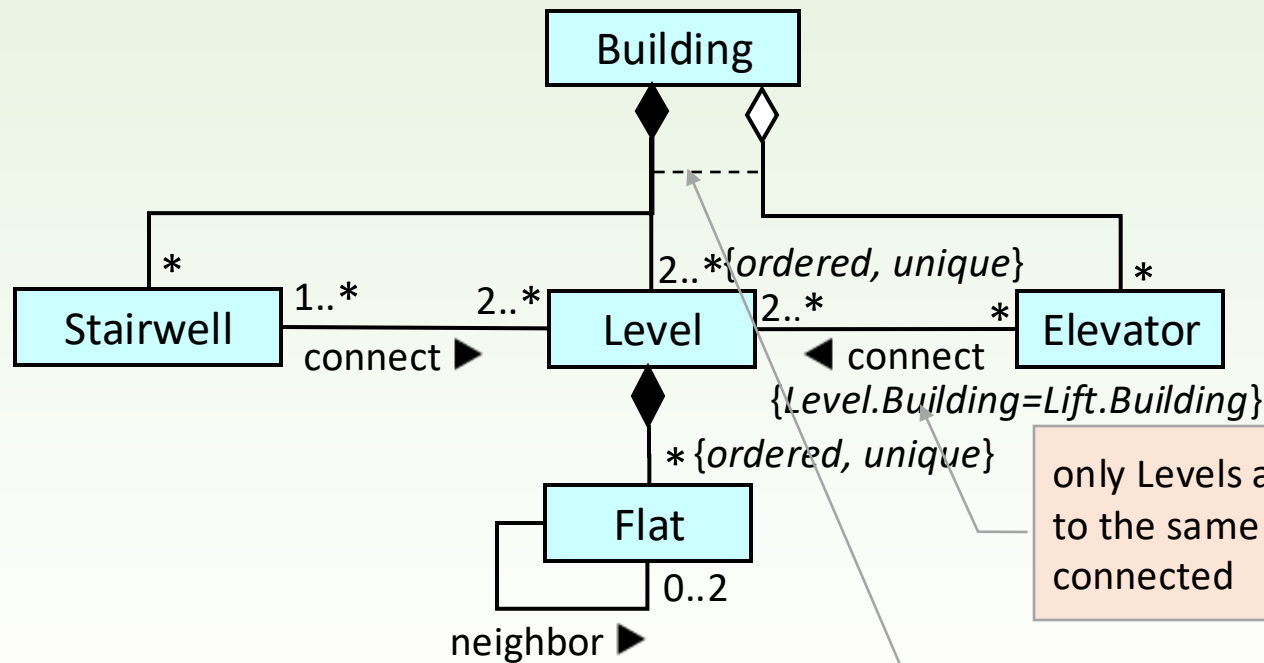
A two-story, five-apartment building without an elevator



A high-rise building containing one luxury apartment per floor



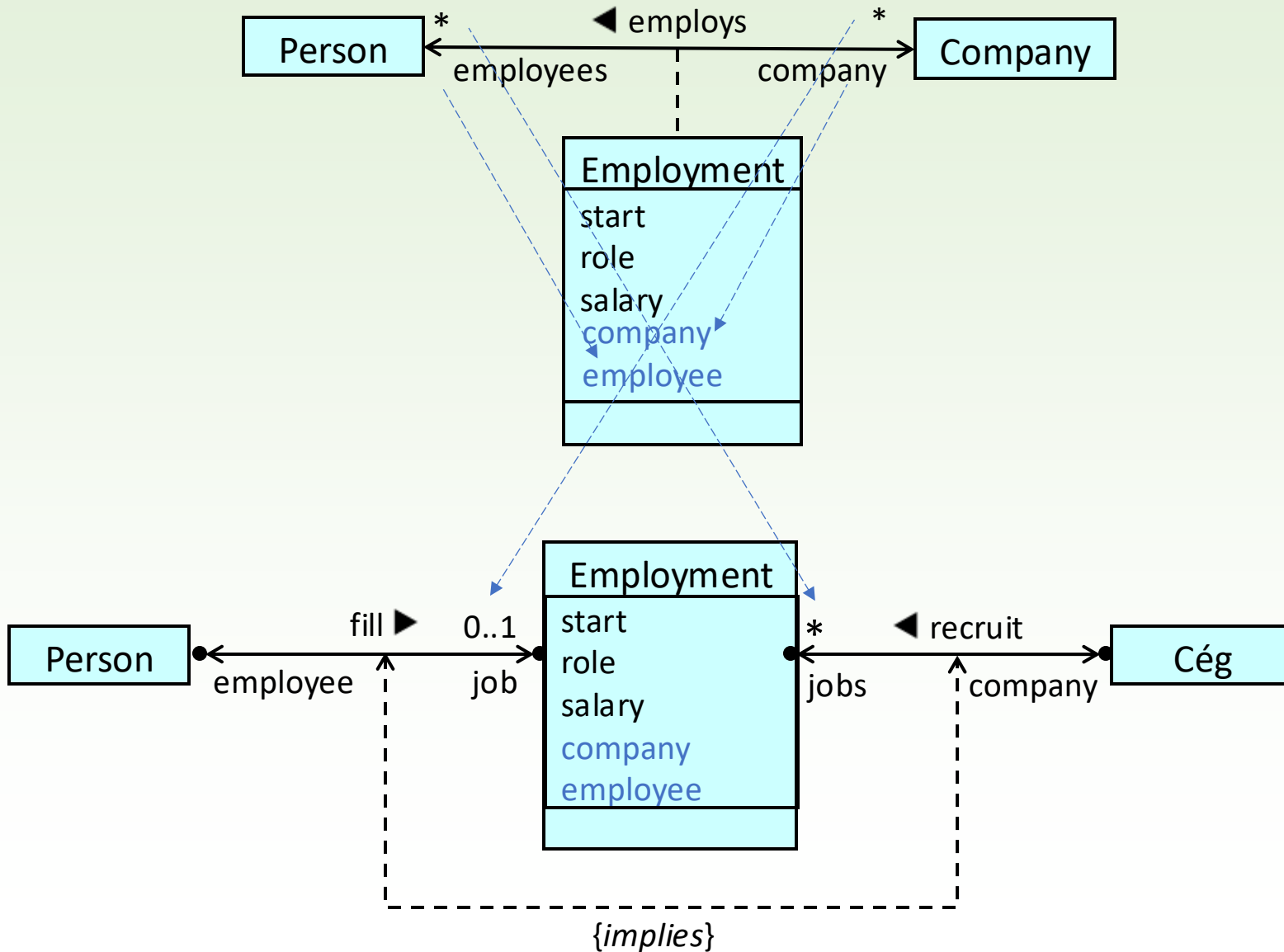
5. A residential building consists of at least two levels, with at least one apartment (flat) per level. The levels are connected by stairwells and elevators, with each level having at least one stairwell. Each apartment can have no more than two neighboring apartments.



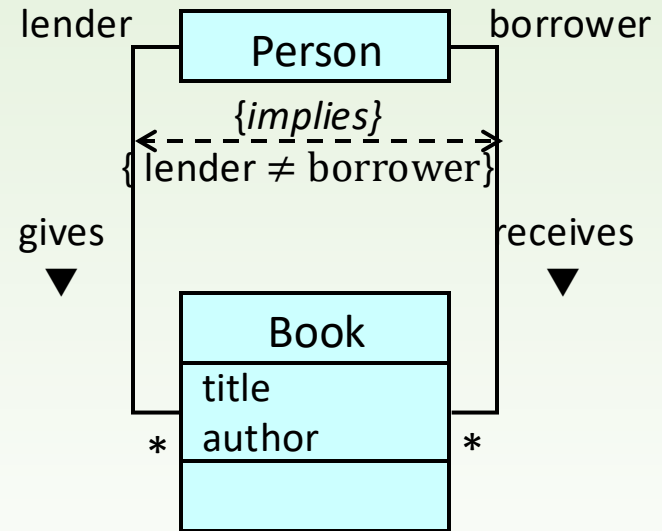
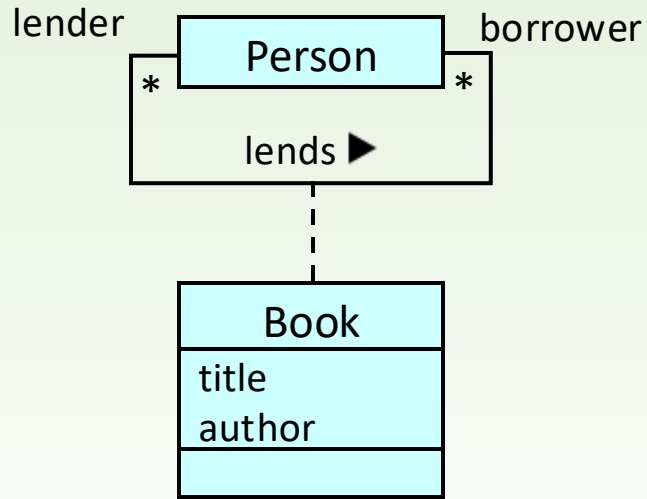
only Levels and Elevators belonging to the same Building can be connected

we could also express it like this, but this is not standard either

6. A company employs individuals in various positions. The job description includes the job start date, the job title, and the salary.



7. One person lends another person a book whose title and author we are familiar with.



8. A company's employees have name, address, and social security number. They are made up of a boss and those reporting to them, where the boss oversees the work of the subordinates. This management involves solving tasks by their deadlines.

